

CHALLENGES AND SOLUTIONS REPORT

Raritone Summer Internship 2026

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Introduction

During the internship, various research and exploration activities were carried out related to Artificial Intelligence, Computer Vision, Virtual Try-On systems, and 3D Avatar Generation. While working on these tasks, several technical and learning challenges were encountered. This report summarizes the major challenges faced during the internship and the approaches used to overcome them.

Challenge 1: Understanding Virtual Try-On Technologies

Problem

At the beginning of the internship, I had limited knowledge about Virtual Try-On systems and the technologies involved in creating realistic digital clothing experiences. Understanding how body detection, pose estimation, segmentation, and garment fitting work together was initially challenging.

Solution

To overcome this challenge, I studied research materials, explored workflow diagrams, and analyzed different Virtual Try-On architectures. Gradually, I developed a better understanding of how these technologies interact to generate virtual clothing visualizations.

Outcome

I gained a clear understanding of the complete Virtual Try-On workflow and its major components.

Challenge 2: Understanding Advanced AI Models

Problem

Several technologies assigned during the internship, such as SMPL, PIFuHD, ICON, HumanNeRF, and 3D Gaussian Splatting, involve advanced concepts in Computer Vision and 3D reconstruction. Understanding their architectures and applications required significant effort.

Solution

I reviewed technical documentation, research articles, comparison studies, and architecture diagrams to understand how each model works and where it is used.

Outcome

I developed a strong conceptual understanding of modern avatar generation and human body reconstruction techniques.

Challenge 3: Dataset Collection and Organization

Problem

Collecting and organizing datasets from multiple sources became challenging due to the large number of images, assets, and reference materials involved.

Solution

A structured folder organization system was created to categorize datasets into clothing images, accessories, footwear, avatars, and 3D assets.

Outcome

The dataset repository became easier to manage and can support future Virtual Try-On research activities.

Challenge 4: Exploring Blender for the First Time

Problem

Blender was a completely new tool, and understanding the interface, navigation controls, and avatar visualization workflow was initially difficult.

Solution

Hands-on exploration was performed to understand Blender's workspace, viewport controls, asset import process, and avatar visualization features.

Outcome

I successfully imported and explored avatar models, understood mesh visualization concepts, and gained confidence working with 3D assets.

Challenge 5: Understanding Human Body Reconstruction

Problem

Human body reconstruction involves complex concepts such as mesh generation, neural rendering, body parameter estimation, and 3D scene reconstruction.

Solution

I compared multiple reconstruction techniques and studied their workflows individually instead of attempting to understand all technologies at once.

Outcome

This approach improved my understanding of how modern AI systems create realistic digital human representations.

Challenge 6: Limited Hardware Resources

Problem

Many advanced reconstruction frameworks require high-performance GPUs, large datasets, and extensive computational resources, making practical implementation difficult within the internship duration.

Solution

Instead of focusing solely on implementation, emphasis was placed on comparative research, workflow analysis, architecture studies, and practical exploration using available tools such as Blender.

Outcome

I gained valuable knowledge about these technologies without being limited by hardware constraints.

Challenge 7: Managing Multiple Research Topics

Problem

The internship covered a wide range of topics including pose estimation, segmentation, garment fitting, avatar generation, body reconstruction, recommendation systems, and AR/VR technologies. Managing multiple research areas simultaneously was challenging.

Solution

Tasks were divided into smaller objectives and completed systematically. Documentation and notes were maintained throughout the internship to track progress.

Outcome

This improved time management and helped ensure that all assigned tasks were completed effectively.

Key Solutions Applied Throughout the Internship

The following strategies were consistently used to overcome challenges:

- Independent learning and research.
- Comparative analysis of multiple technologies.
- Workflow-based understanding.
- Structured documentation.
- Asset organization and management.
- Practical exploration using Blender.
- Continuous review of research materials.

Lessons Learned

The challenges encountered during the internship became valuable learning opportunities. They improved my problem-solving abilities, research skills, technical understanding, and confidence in exploring unfamiliar technologies.

The experience demonstrated the importance of continuous learning, adaptability, and structured research when working with emerging technologies.

Conclusion

Although several technical and learning challenges were encountered during the internship, each challenge provided an opportunity to acquire new knowledge and develop practical skills. Through research, exploration, and documentation, I successfully gained a deeper understanding of Artificial Intelligence, Computer Vision, Virtual Try-On systems, and 3D Avatar Generation technologies. The solutions applied throughout the internship helped transform challenges into valuable learning experiences.